

Satyajith Chilappagari

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WORK EXPERIENCE

AMAZON.COM, INC.

CUPERTINO, CA, USA

Machine Learning Engineer II at Annapurna Labs

July 2024 - Present

- **Owned async scheduling on AWS Neuron end to end**, from design through implementation and validation, eliminating per-token CPU-device round trips and cutting TPOT by **20–50%** across models. [[Design doc](#)]
- **Optimized EAGLE3 speculative decoding draft models**, adding data-parallel support for the LM head, embedding, and MLP layers. Built a fused kernel for the GPT-OSS draft model, cutting device execution time by **~47%**.
- **Extended vLLM support for Trainium and Inferentia**, implementing capabilities for standard inference, multimodal inference, speculative decoding, quantization, and multi-LoRA using the AWS Neuron SDK. [[merged PRs](#)]
- **Built automated LLM benchmarking and regression pipelines** integrating vLLM and LLMPerf, cutting developer benchmarking time by **90%+** and validation cycles from **6 weeks to 1** through CI/CD-gated weekly reviews.
- **Implemented isolated prefill and decode benchmarking for disaggregated inference**, enabling org-wide separate prioritization of prefill and decode work and more transparent reporting of results.
- **Designed and implemented an upstreaming workflow** to contribute AWS Neuron vLLM enhancements from internal packages to the open-source vLLM repository and vLLM-Neuron plugin, cutting release timelines from **60+ days to 2 days**. Personally shipped 7 releases from 0.2.0 to 0.5.3. [[Releases](#)]

Software Development Engineer at Alexa

Feb 2022 - July 2024

- **Developed a real-time ingestion platform for Alexa Proactive Experiences** (30K+ TPS) supporting validation, transformation, and dispatch. Downstream datasets powered product analytics via Athena.
- **Designed and integrated a version-supported schema validation module**, ensuring uniform event structure across all Alexa proactive interactions.
- **Implemented pipeline-based staged rollout in production environments**, reducing impact of faulty deployments to just 1% of traffic. This contained a database change that would have caused a 3-day data loss.
- **Reduced costs by 40% per month** by replacing AWS Firehose with a custom module that batches and writes data in Parquet format to S3.

DRESSLIFE GMBH

HANNOVER, GERMANY

Backend Developer

Mar 2017 - Dec 2019

- **Engineered the back-end of a fashion-based mobile application** using an event-driven architecture based on Apache Kafka, Cassandra and Elasticsearch.
 - **Developed a size recommendation module** leveraging customers' height and weight data to simulate bodies.
 - **Built a personalized style recommendation engine** using autoencoders and convolutional neural networks (CNNs) to reduce body and clothing features into embeddings and perform nearest-neighbor retrieval in the learned space.
 - **Deployed the system in production for two German clothing brands**, driving a 5% revenue increase by improving fit accuracy and boosting conversion through more personalized suggestions.
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EDUCATION

RUTGERS, THE STATE UNIVERSITY OF NEW JERSEY

NEW BRUNSWICK, NJ, USA

MS in Computer Science (Specialization: Machine Learning) – GPA 3.90/4

Jan 2020 - Jan 2022

- **Part-time lecturer and grader**: Taught a university level Data 101 course using R as the primary language.
- **Internship at Commvault (June 2021 - December 2021)**: Secured second place among 70 interns in the intern project competition for developing an NLP-based bug-search tool.

SKILLS

LLM Inference: vLLM, speculative decoding (EAGLE3), disaggregated prefill/decode, async scheduling, quantization, multi-LoRA, multimodal inference, data parallelism, LLM benchmarking (LLMPerf)

ML & Hardware: PyTorch, AWS Neuron SDK, Trainium, Inferentia, custom kernel development, CNNs, autoencoders

Languages: Python, Java, C++, TypeScript

Data, Cloud & DevOps: Kafka, Cassandra, Elasticsearch, DynamoDB, SQL, AWS (EC2, S3, Lambda, Kinesis, Athena), Docker, CI/CD, Grafana